Training-Free Partial-Order Control for Discrete-Diffusion POI Check-in Generation

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Abstract

Location-based social networks (LBSNs) are a core source of web-scale human-mobility data, yet sharing raw check-in traces is privacy-sensitive, motivating realistic synthetic generation. Beyond statistical realism, many downstream web applications need control over the *semantic order* of activities—e.g., leisure before dining—a sample-level, cross-position requirement that single-step conditioning cannot express. Enforcing such constraints is commonly assumed to cost distributional fidelity; we show this trade-off is largely *separable*. We present PCDG, a *training-free* control stack for a frozen conditional joint temporal + category/POI discrete-diffusion generator. PCDG compiles category partial-order constraints into differentiable energies, projects the category-position logits through a KL-preserving augmented-Lagrangian optimization in the late reverse steps, eliminates residual category–POI inconsistency via a category-consistent decoder, and restores spatial realism through a distribution-aware same-category geometry refinement that leaves every constraint metric unchanged *by construction*. Across two cities under an out-of-distribution split and three sampling seeds, PCDG cuts the strict ordering-violation rate from 92.89% to 4.04% on Istanbul and from 89.45% to 11.28% on New York, raising pairwise ordering coverage by 6.5× and 4.2×, while *improving* all six fidelity divergences over the unconstrained backbone on Istanbul and four of six on New York. On Istanbul it surpasses a *fully retrained* partial-order conditional model on all five constraint metrics and five of six fidelity divergences, without retraining anything. Stage-wise ablations localize where fidelity is lost and recovered across the three-stage stack.

CCS Concepts

• **Computing methodologies** → **Machine learning**; • **Information systems** → *Information systems applications.*

Keywords

synthetic human mobility, discrete diffusion, partial-order constraints, training-free constrained sampling, POI check-in sequences

1 Introduction

Location-based social networks (LBSNs) and location-aware web services generate vast streams of *check-in* data: time-stamped visits to points of interest (POIs), each carrying an activity semantics such as dining, leisure, or shopping. These web-scale traces underpin POI recommendation, urban analytics, and mobility simulation [9, 26, 41]. Yet releasing raw traces is privacy-sensitive, and long-tail scenarios are scarce, which has made the generation of realistic *synthetic* mobility an active research direction [7, 10, 24, 43].

Most prior generators optimize statistical realism alone. Many web applications, however, require *control over activity semantics*—in particular, over the *order* in which categories of activities occur. A scenario designer may need sequences in which leisure precedes dining, or sightseeing precedes shopping, in order to study specific behavioral patterns, stress-test a recommender, or augment data for a target scenario. Such a requirement is a *sample-level partial order* over categories: it couples multiple, possibly non-adjacent positions of a sequence and therefore cannot be expressed by reweighting any single token, nor by next-location prediction, which outputs one step at a time rather than a complete constrained sequence.

Enforcing such constraints is surprisingly hard for modern generators. Generating freely and then *reordering* the output satisfies the constraint but distorts the jointly learned distribution over time, category, and space. Classifier-free or classifier-based guidance can bias generation toward a condition, but a condition embedding does not *guarantee* a global ordering, and retraining a conditional model for every constraint family is expensive and city-specific. Recent work on controllable trajectory generation enforces *geometric* constraints such as road topology [46], and generic constrained discrete diffusion enforces sequence-level logical constraints [3]; neither has been specialized to the activity-category precedence that arises in POI check-in sequences, nor to the joint temporal + category + POI setting.

We introduce PCDG, a *training-free* control stack that enforces sample-level category precedence directly inside the reverse diffusion of a frozen conditional generator, and that—crucially—does so without paying the fidelity penalty such control is usually assumed to require. PCDG has three sampling-time stages. First, it compiles a partial-order matrix into differentiable order-violation and existence energies and, during the late reverse steps, projects *only the category-position logits* via a KL-preserving augmented-Lagrangian optimization. Second, a category-consistent POI decoder resolves the residual mismatch between a sampled category and the POI actually emitted at that position. Third, a distribution-aware same-category geometry refinement re-selects each POI among candidates of the *same* realized category so as to match population-level travel-length and visitation statistics; because it never touches category tokens, times, lengths or order, it leaves every constraint metric unchanged by construction. Because all three stages read and rewrite only the per-step sampling distribution and the realized POI assignment of a frozen model, PCDG adds no parameters, needs no retraining, and can be attached to an existing, already-deployed generator.

Our analysis traces the fidelity cost of constrained sampling to two separable sources—an over-weighted constraint penalty inside the projection, and a POI/coordinate selection that is blind to the population spatial distribution—and shows that correcting them recovers, and on one city surpasses, the unconstrained generator's fidelity while retaining almost all of the constraint gain.

Relevance to the Web. This work studies the Web as a socio-economic system and a responsible-data problem. LBSN check-in logs are a native web data source that powers POI recommendation

and urban web services, but releasing raw traces risks user privacy. Generating controllable, constraint-satisfying synthetic mobility lets practitioners share and augment web-behavior data while honoring application-specified activity semantics, directly serving privacy-conscious web data sharing and downstream recommendation. The method also contributes a reusable recipe—training-free, sampling-time constraint projection on a frozen generator—for controllable generation of other structured web content.

Contributions.

- We formalize *category partial-order constrained POI check-in generation*—a sample-level, cross-position precedence requirement with an anti-degeneracy existence condition—and express it as differentiable energies compatible with discrete-diffusion sampling (§3, §4.2).
- We propose PCDG, a three-stage *training-free* sampling-time control stack on a *frozen* joint temporal + category + POI diffusion generator: a KL-preserving augmented-Lagrangian projection of the category-position logits, category-consistent POI decoding, and a distribution-aware same-category geometry refinement that is constraint-neutral by construction (§4).
- We show that the fidelity–constraint trade-off reported for constrained sampling is *separable* rather than intrinsic: it decomposes into an over-weighted constraint penalty and a distribution-blind POI/coordinate selection. Correcting both recovers the unconstrained generator's fidelity on every measured divergence on one city, and the configuration that wins uses *no* category-side distribution term, localizing the loss to the spatial side (§5.4).
- We conduct a two-city, out-of-distribution, multi-seed study with a projection ablation and a stage-wise ablation of the full stack, showing that PCDG attains the strongest constraint satisfaction among energy-guidance, post-hoc, and even *fully retrained* conditional baselines while matching or beating them on fidelity, and we report the two New York statistics that remain above the unconstrained backbone (§5).

2 Related Work

Human mobility and POI modeling. Classical studies established that human movement is highly regular yet only boundedly predictable [13, 34]. A large body of work predicts the *next* location from history using attentional, spatio-temporal, and graph-enhanced models [9, 26, 41], and diffusion models have recently been applied to next-POI and sequential recommendation [6, 22, 29, 36, 38]. These methods target prediction or ranking of a single next step and do not emit a complete, constraint-satisfying sequence of time-stamped, categorized visits.

Synthetic trajectory and mobility generation. Data-driven generation spans GAN/Seq2Seq models [2, 19, 21, 40], point-process and neural-ODE generators [24, 43], imitation- and privacy-oriented approaches [10, 17, 28, 30, 35, 37], and, increasingly, diffusion models [4, 12, 23, 31, 39, 42, 45] and LLM-based generators [14, 16, 20, 44]. Closest to our backbone, Marionette performs fine-grained conditional generation of spatiotemporal trajectories with a diffusion point process and a category-to-POI cascade [7], and related benchmarks study imitative generation quality [8]. These works pursue distributional realism or coarse conditioning; none enforces a *sample-level partial order over activity categories* within the generated sequence.

Discrete diffusion. Discrete diffusion extends denoising diffusion [15] to categorical data via structured transition matrices [1], concrete score matching [27], score-entropy [25], and masked formulations [32]; guidance mechanisms steer discrete generation toward conditions [33]. We adopt a discrete-diffusion backbone and intervene at sampling time rather than changing training.

Controllable and constrained generation. ControlTraj injects road-topology constraints into trajectory diffusion [46]; Geo-Llama conditions an LLM on mobility constraints [20]. In NLP, structured decoding enforces grammars or DAG automata [5, 11], and precedence constraints are studied in temporal planning [18]. Most relevant methodologically, Constrained Discrete Diffusion (CDD) embeds differentiable constraint optimization into discrete-diffusion sampling via KL projection, Gumbel-softmax relaxation, and augmented Lagrangian updates [3]. PCDG *specializes* this sampling-time projection paradigm to the previously unaddressed case of activity-category precedence in joint temporal + category + POI check-in generation: we contribute the order/existence energy formulation for sample-level partial orders, a category-position-restricted projection coupled with a spatial distance-distribution regularizer on POI positions, and—unlike prior controllable generators that retrain or post-process—a training-free controller that we show outperforms a *fully retrained* partial-order conditional model on constraint satisfaction (§5).

3 Problem Formulation

POI check-in sequences. We model a user's daily behavior on an LBSN as a variable-length sequence of check-in events. A sequence of length n consists of arrival times $t_{1:n}$ (continuous, strictly increasing), activity categories $k_{1:n}$ with $k_i \in \mathcal{C} = \{1, \ldots, C\}$, and points of interest $l_{1:n}$ with $l_i \in \mathcal{L}$. Each POI l carries fixed geographic coordinates $\mathrm{gps}(l)$, and each training POI is associated with a category. A per-sequence context $\mathbf{c}$ encodes coarse temporal and behavioral conditions (e.g., time-of-day occupancy indicators). The empirical distribution over such sequences is highly structured: it reflects the regularity and bounded predictability of human mobility [13, 34] while retaining substantial individual variability.

Category partial-order constraints. Many web applications do not merely require a statistically realistic sequence; they require that the *activity semantics* follow an application-specified order—for instance, that leisure precedes dining, or that sightseeing precedes shopping. We formalize such requirements as a sample-level *partial order* over categories. Given a reference or user-specified ordering, we encode it as a binary matrix $\mathbf{M} \in \{0,1\}^{C\times C}$, where $\mathbf{M}[a,b] = 1$ denotes the strict precedence constraint "category a precedes category b": in a satisfying sequence, *every* occurrence of category a must appear before *every* occurrence of category b. When $\mathbf{M}$ is derived from a reference sequence, we set $\mathbf{M}[a,b] = 1$ iff the last position of a precedes the first position of b. A constraint (a,b) is *violated* either by an inversion (some b before some a) or, degenerately, by dropping a or b entirely so that the ordering is vacuously "satisfied." A faithful method must avoid both.

Task. Given a frozen conditional generator of complete check-in sequences and a sample-level partial-order matrix $\mathbf{M}$, we seek to sample

$$(t_{1:n},\, k_{1:n},\, l_{1:n}) \sim p\big(t_{1:n}, k_{1:n}, l_{1:n} \mid \mathbf{c}, \mathbf{M}\big), \tag{1}$$

such that (i) the generated sequences remain close to the real data distribution in time, space, and activity semantics; and (ii) the category subsequence satisfies the precedence constraints encoded by $\mathbf{M}$. Unlike next-location prediction [9, 26, 41], the task outputs a *complete* time-stamped, categorized, geolocated sequence; unlike single-step conditioning, the constraint couples *multiple, non-adjacent* positions and therefore cannot be enforced by reweighting any single token independently.

Evaluation axes. We evaluate along two axes that are in tension. *Distributional fidelity* is measured by the Jensen–Shannon divergence (JSD) between generated and reference marginals of travel distance, radius of gyration, category-transition structure, daily location count, hourly category usage, and a global ranking statistic (G-RANK). *Constraint satisfaction* is measured by ordering-violation rates (OVR), pairwise and category coverage of the required relations, and an overall unsatisfied-sequence rate (Unsat); full definitions are given in Section 5. A trivial reordering can satisfy the constraint while destroying fidelity, and an unconstrained generator maximizes fidelity while ignoring the constraint; the goal is to improve constraint satisfaction with minimal fidelity cost.

4 Method

PCDG controls a frozen generator at sampling time through logit projection, category-consistent POI decoding, and terminal same-category geometry refinement. Figure 1 combines the framework and a controlled reverse step; ρ, λ, β enter the projection and refinement objectives, rather than forming a separate execution stage.

4.1 Backbone: conditional joint discrete diffusion

We build on a conditional generator that models arrival times with a diffusion point process and the discrete content (categories and POIs) with a conditional discrete diffusion model, following the fine-grained trajectory generator of Deng et al. [7]. The temporal branch samples event times; the discrete branch represents a sequence as a token string $\mathbf{x} = [\textsc{start}, k_{1:n}, \textsc{sep}, l_{1:n}, \textsc{end}]$, with a *category mask* and a *POI mask* identifying the two segments. Categories and POIs use separate absorbing-state transition matrices [1]; a Transformer denoiser with an $\mathbf{x}_0$-parameterization predicts, at each reverse step, per-position logits $\mathbf{z} \in \mathbb{R}^{V \times L}$ over the joint vocabulary of size V across the L positions, from which the posterior $p(\mathbf{x}_{s-1} \mid \mathbf{x}_s)$ and the next discrete state are sampled. **The backbone is used as-is and is never retrained by PCDG**; our contribution modifies only the per-step sampling distribution $\mathbf{z}$.

4.2 Partial-order constraints as differentiable energies

Given the sample-level matrix $\mathbf{M}$, we parse it into a set of pairwise constraints $\{(a,b) : \mathbf{M}[a,b] > \frac{1}{2}\}$. To make a constraint differentiable in the logits, we relax the hard categorical choice at each position into a distribution. With temperature T and (optionally) Gumbel noise $\mathbf{g}_l$,

$$\boldsymbol{\pi}_l = \mathrm{softmax}\big((\mathbf{z}_l + \mathbf{g}_l)/T\big) \in \Delta^{V-1}, \qquad l = 1, \ldots, L. \tag{2}$$

We restrict attention to category positions $\mathcal{P}_k$ (via the category mask) and to the category slice of the vocabulary, writing $\pi_l^{\mathrm{cat}}(a)$ for the relaxed probability that position $l \in \mathcal{P}_k$ takes category $a \in C$. For a constraint (a,b) let $P_l^A = \pi_l^{\mathrm{cat}}(a)$ and $P_l^B = \pi_l^{\mathrm{cat}}(b)$. We define two energies:

$$g^{\mathrm{ord}}_{(a,b)} = \sum_{l \in \mathcal{P}_k} \Big(\textstyle\sum_{l' < l} P^B_{l'} \Big) P^A_l, \tag{3}$$

$$g^{\mathrm{ex}}_{(a,b)} = \mathrm{ReLU}\Big(1 - \textstyle\sum_l P^A_l\Big) + \mathrm{ReLU}\Big(1 - \textstyle\sum_l P^B_l\Big). \tag{4}$$

The *order energy* (3) is the expected number of inverted pairs, i.e. the probability mass of a b appearing *before* an a; it is zero iff no b precedes any a. The *existence energy* (4) penalizes an expected count below one for either constrained category, which prevents the model from trivially satisfying ordering by *omitting* a or b. Both energies are computed only over category positions; POI logits do not enter Eqs. (3)–(4).

4.3 KL-preserving augmented-Lagrangian projection

At a controlled reverse step, we project the backbone logits to reduce constraint violations while staying close to the model's own distribution. Let $\mathbf{y}^{(0)} = \mathbf{z}$ be the detached backbone logits and $\mathbf{y}$ a trainable copy. We solve, per sequence,

$$\begin{aligned} &\min_{\mathbf{y}} D_{\mathrm{KL}}\big(\mathrm{softmax}(\mathbf{y}) \,\big\|\, \mathrm{softmax}(\mathbf{y}^{(0)})\big) \\ &\text{s.t. } g^{\mathrm{ord}}_{(a,b)} \le \tau, \quad g^{\mathrm{ex}}_{(a,b)} \le \tau \quad \forall (a,b), \end{aligned} \tag{5}$$

where the KL term anchors the projected distribution to the base model so that fidelity is preserved away from the constrained categories. We optimize the augmented-Lagrangian relaxation of (5): with $\delta^\bullet = \mathrm{ReLU}(g^\bullet - \tau)$,

$$\begin{aligned} \mathcal{L}(\mathbf{y}) = w_{\mathrm{kl}}\, D_{\mathrm{KL}} &+ \sum_{(a,b)} w_{\mathrm{ord}} \Big(\lambda^{\mathrm{ord}} \delta^{\mathrm{ord}} + \tfrac{1}{2} \mu^{\mathrm{ord}} (\delta^{\mathrm{ord}})^2 \Big) \\ &+ \sum_{(a,b)} w_{\mathrm{ex}} \Big(\lambda^{\mathrm{ex}} \delta^{\mathrm{ex}} + \tfrac{1}{2} \mu^{\mathrm{ex}} (\delta^{\mathrm{ex}})^2 \Big). \end{aligned} \tag{6}$$

An *inner* loop takes projected-gradient steps on $\mathbf{y}$ (SGD with learning rate η, gradient-norm clipping, updates restricted to positions of constrained sequences). An *outer* loop evaluates the *hard* violations $\hat{\delta}$ (from the arg max one-hot assignment, not the relaxation) and updates the dual variables,

$$\lambda \leftarrow \lambda + \mu\, \hat{\delta}, \qquad \mu \leftarrow \min(\alpha\, \mu,\ \mu_{\max}) \text{ if } \hat{\delta} > \tau_{\mathrm{tol}}, \tag{7}$$

stopping early once $\max \hat{\delta} < \tau_{\mathrm{tol}}$. Evaluating the multiplier update on hard violations lets the penalty grow precisely for the sequences and constraints that remain unsatisfied, rather than applying a uniform fixed penalty. Crucially, the projection reweights the sampling distribution over category positions only; it performs *no* POI candidate filtering and imposes *no* hard category→POI consistency mask.

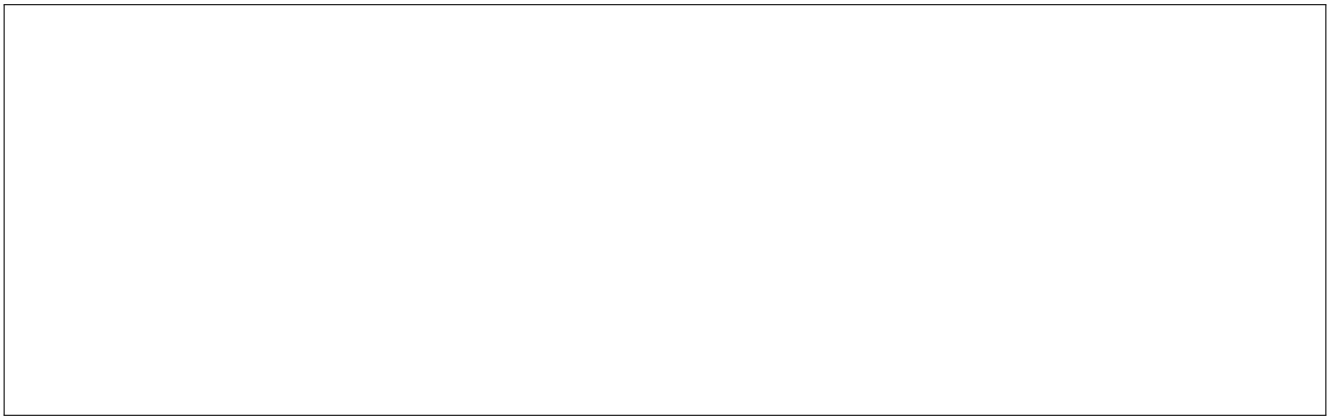


Figure 1: Framework and one controlled reverse step (reserved layout for T3).

4.4 Distance-distribution regularization

Enforcing category order perturbs the category marginals and, indirectly, the induced spatial statistics. To preserve geographic realism we add an independent distance regularizer that matches the *population* distribution of travel length. From the training split only, we compute the cumulative great-circle (Haversine) route length d of every training sequence with at least two POIs and summarize it as a soft histogram H_{train} over $\log(1+d)$ with fixed Gaussian bins (the reference is content-hashed and cached, never stored in the model). During projection, for each POI position we fix the top-κ candidate POIs under the backbone and draw P routes with a straight-through Gumbel-softmax (exact distance on the forward pass, a bilinear surrogate on the backward pass), forming a differentiable route-length histogram $H_{\text{proj}}(\mathbf{y})$. We add the term

$$w_{\text{dist}}\, D_{\text{KL}}\big(H_{\text{train}} \,\|\, H_{\text{proj}}(\mathbf{y})\big) \tag{8}$$

to the objective (6). Unlike the order/existence energies, this term acts on *POI-position* logits, and it is a batch-level distribution-matching regularizer rather than a per-sample hard constraint; it is applied only to sequences with at least two POIs and uses an independent random stream. Its weight is not tuned per city and is not guaranteed to improve every fidelity metric; we quantify its effect via the No Distance KL ablation (§5).

4.5 Placement in the reverse process and joint sampling

The control is applied in the *late* reverse-diffusion steps—where the predicted categories are informative enough for the energies to be meaningful—specifically during the last K steps and every f steps therein. At each such step, PCDG replaces the backbone category-position logits with their projected values (and, when enabled, the distance-adjusted POI-position logits), and the next discrete state is obtained by the backbone's ordinary joint categorical sampling over the full token sequence, optionally through the category-consistent decoder of §4.6. The temporal branch is unchanged. Because every operation reads and rewrites only the per-step sampling logits of a frozen model, PCDG adds no trainable parameters and requires no retraining, and it reduces to the unconstrained backbone when the projection is disabled. Algorithm pseudocode and default hyperparameters are given in Appendix A and Figure 1.

4.6 Category-consistent POI decoding

The projection of §4.3 acts on category positions, but the backbone emits categories and POIs as separate segments of the same token string and samples every position from its own categorical. Nothing ties the POI emitted at position l to the category sampled at position l. This matters because the constraint metrics are evaluated on the *actual* category of each generated POI, $\text{cat}(\cdot)$ from the POI catalog, not on the category tokens: a sequence whose category tokens are perfectly ordered still violates $\mathbf{M}$ if the POIs realize a different category sequence. On a 128-sequence single-seed diagnostic (not one of the formal runs) the projected backbone left 17.5% of positions category–POI inconsistent; enabling the decoder below drove that rate to 0 and, *without altering a single category token*, cut the strict violation rate from 27.0% to 2.2%—so almost all of those violations came from the POI segment disagreeing with an already correctly ordered category segment.

We therefore decode the two segments *sequentially within one step* rather than independently. Let $\mathbf{g}$ be the Gumbel draw used by the backbone's categorical sampler at this step. We first take the ordinary sample $\hat{x}_l = \arg\max_v (\mathbf{g}+\mathbf{z})_{lv}$, which fixes the realized category at every category position. We then set, at each POI position (the set $\mathcal{P}_{\text{poi}}$ given by the POI mask) the logits of every POI whose catalog category differs from that realized category to $-\infty$ and re-take the arg-max *reusing the same draw* $\mathbf{g}$:

$$\hat{x}_l^{\text{cc}} = \arg\max_v \big(\mathbf{g}+\widetilde{\mathbf{z}}\big)_{lv}, \quad \widetilde{z}_{lv} = \begin{cases} -\infty & l \in \mathcal{P}_{\text{poi}},\ \text{cat}(v) \neq \hat{c}_l \\ z_{lv} & \text{otherwise,} \end{cases} \tag{9}$$

where $\hat{c}_l$ is the category realized for the event at position l. Three properties follow. (i) Category positions are never rewritten, so the realized category sequence—and hence the ordering the projection just enforced—is preserved exactly. (ii) Exactly one draw is consumed per step, identical in shape and order to the unmodified sampler, so the "off" and "on" runs are paired at matched randomness and no extra global, spatial, projection or distance randomness is used. (iii) Masking uses true $-\infty$ rather than a large finite penalty, which would let a low-scoring legal POI tie with an illegal one. At

intermediate reverse steps the POI mask state is still admissible and no restriction is applied when the category is itself masked or no legal POI exists; at the final step only same-category POIs are legal and the sampler raises rather than silently emitting an inconsistent token. The decoder adds no parameters and no logit prior—it is a restriction on realized tokens.

4.7 Terminal same-category geometry refinement

The stages so far decide *which* category and *which* POI identity; the coordinates are then whatever the chosen POIs happen to be, and the spatial marginals drift as a result (§5.4). After the reverse process completes, we therefore re-select each POI *within its own realized category*.

For every POI position we build a candidate set of at most κ_g=32 POIs sharing the realized category: the originally sampled POI, the backbone's highest-scoring candidates, the POIs closest to the generated trajectory's own centroid, and—in the distribution-corrected variant—the most frequently visited POIs of the training reference; duplicates are removed, the set is filled by model score, and ties are broken by catalog id so the construction is deterministic. Scores come from one frozen, untruncated t=0 evaluation of the backbone; reference GPS from the test split never enters. Writing $\mathbf{w}$ for a relaxed assignment over candidates, we minimize

$$D_{\mathrm{KL}}\big(q \,\|\, \mathbf{w}\big)\, w_{\mathrm{fid}} + w_{\mathrm{d}}\, \mathrm{JSD}\big(H^{\mathrm{dist}}_{\mathrm{train}}, H^{\mathrm{dist}}(\mathbf{w})\big) + w_{\mathrm{r}}\, \mathrm{JSD}\big(H^{\mathrm{rad}}_{\mathrm{train}}, H^{\mathrm{rad}}(\mathbf{w})\big), \tag{10}$$

where the two soft histograms are differentiable in $\mathbf{w}$ and the proposal q places half of its mass on the original POI and spreads the other half over the candidates by model score, so the fidelity term penalizes drifting away from the backbone's own preference. We optimize Eq. (10) with P_g=8 straight-through relaxed paths, temperature 1, Adam at learning rate 0.05, for a prequalified budget of 200 steps, refreshing an independent geometry noise stream every 20 steps. The output is a *hard* assignment: among the original assignment and the 8 optimized paths we keep the arg-min of the same training-side objective. Empty, single-point and single-candidate positions, and batches in which the base projection never ran, are skipped.

Constraint neutrality. Because the refinement substitutes only POIs of the same realized category, together with their catalog coordinates, and leaves category tokens, the realized category sequence, event times, sequence lengths, event order and all conditions unchanged *per event*, every constraint-relevant statistic is invariant by construction: $\mathrm{OVR}_{\mathrm{strict}}$, $\mathrm{OVR}_{\mathrm{skip}}$, pair and category coverage, Unsat, *Category*, *CategoryTransition* and the category–POI mismatch rate. We verify this numerically on every test run, requiring agreement within 10^{-12} rather than assuming it. The refinement can therefore only move the spatial and popularity statistics—which is precisely why it can repay the fidelity cost of the constraint stages without giving any of their gains back.

4.8 Distribution correction

Three scalars govern the balance between constraint strength and distributional fidelity, all referenced to the *training* reference pool only. First, $\rho \in (0, 1]$ scales the constraint term of Eq. (6), so the projection objective becomes

$$w_{\mathrm{kl}}\, D_{\mathrm{KL}} + \rho \sum_{(a,b)} \mathrm{AL}\big(g^{\mathrm{ord}}, g^{\mathrm{ex}}; \lambda^{\bullet}, \mu^{\bullet}\big) + w_{\mathrm{dist}}\, D_{\mathrm{KL}}\big(H_{\mathrm{train}} \| H_{\mathrm{proj}}\big) + \lambda\, \mathcal{L}_{\mathrm{cat}}. \tag{11}$$

Second, $\lambda \geq 0$ weights a category-side term

$$\mathcal{L}_{\mathrm{cat}} = \tfrac{1}{2}\big(\mathrm{JSD}_{\mathrm{hour}\times\mathrm{cat}} + \mathrm{JSD}_{\mathrm{transition}}\big), \tag{12}$$

computed from *expected* counts under the relaxed assignment—no Monte Carlo draws, so no additional randomness is consumed—with self-transitions counted, sequence boundaries excluded, and reference rows without support contributing zero. Third, $\beta \geq 0$ adds to Eq. (10) a POI-side term $\frac{1}{2}(\mathrm{JSD}_{\mathrm{top1000}} + \mathrm{JSD}_{\mathrm{DailyLoc}})$, where the first is an *identity-aligned* top-1000 visitation divergence (the generated and reference top-1000 sets are unioned by POI id before normalization, not compared as sorted popularity curves) and the second a soft expected-*DailyLoc* divergence. Spatial optimization results are cached independently of β, so the three β settings of a batch are compared at identical upstream randomness.

The configuration used throughout is ρ=0.5, λ=0, β=1, selected on a training-side calibration grid (Appendix B). We retain λ in the formulation because it is the natural category-side counterpart of β; that the selected configuration switches it *off* is an empirical finding about where the fidelity cost lives (§5.4), not a modelling assumption.

5 Experiments

We evaluate whether PCDG enforces category precedence while preserving fidelity, how it compares to retraining- and post-processing-based alternatives, and how each component contributes.

5.1 Setup

Datasets. We use check-in data from two cities, **New York** and **Istanbul**, under an out-of-distribution (OOD) split that holds out partial-order configurations unseen during training (2,108 and 4,914 held-out reference sequences, respectively). For each reference we derive a sample-level partial-order matrix $\mathbf{M}$ from the category first/last positions (§3). The base generators are trained once per city and **kept frozen** for all methods; PCDG and all sampling-time baselines reuse the identical frozen weights. [TODO: confirm final train/test counts and OOD definition from distance-v2 manifest]

Protocol. Unless noted, the constraint projection uses order weight w_{ord}=1, existence weight w_{ex}=5, token-KL weight w_{kl}=1, and distance-KL weight w_{dist}=1 (8 routes, top-32 candidates, 32 bins), with Gumbel temperature 3, applied over the last 40 reverse steps every 4 steps without early stopping. No per-city hyper-parameter search is performed. We report the mean ± sample standard deviation over **three sampling seeds** at a fixed checkpoint; the reported variability reflects sampling stochasticity, not training variability.

Baselines. We compare four external methods on the same frozen backbone unless stated: **JointGen** (unconstrained backbone); **PostSwap** (generate, then stably reorder categories to satisfy $\mathbf{M}$); **EnergyGuide** (gradient guidance by the constraint-violation energy during sampling [33]); and **PO-CFG** (a *retrained* classifier-free-guidance model conditioned on the partial order, trained for 1000 epochs; guidance

scale 1). PO-CFG is the only method that requires retraining. We report three variants of our own stack: PCDG-P (projection only), PCDG-PD (adding category-consistent POI decoding), and the full PCDG (adding distribution-aware geometry refinement).

Metrics. Fidelity uses the Jensen–Shannon divergence (JSD, lower is better) of six marginals (evaluation protocol v2): travel *Distance*, *Radius* of gyration, *CategoryTransition* (adjacent directed category pairs), *DailyLoc* (daily location count), hourly *Category* usage, and a global rank statistic *G-RANK*. *Constraint satisfaction* (percentages) uses OVR_{strict} and OVR_{skip} (ordering-violation rates), *Pair coverage* and *Category coverage* (fraction of required relations/categories realized, higher is better), and *Unsat* (fraction of sequences violating any constraint, lower is better). [TODO: add one-line formal definitions in Appendix D]

5.2 Main results

Table 1 reports the method comparison on both cities.

Constraint satisfaction. The full PCDG reduces the strict ordering-violation rate from 92.89% to 4.04% on Istanbul and from 89.45% to 11.28% on New York (−88.85 and −78.17 percentage points; −95.7% and −87.4% relative), lowers *Unsat* from 95.57% to 7.29% and from 93.64% to 15.86%, and raises pairwise ordering coverage by 6.5× and 4.2× respectively. It exceeds the *retrained* PO-CFG on all five constraint metrics in both cities by a wide margin (63.45% → 4.04% and 41.91% → 11.28% strict violations), despite training nothing. PostSwap drives OVR_{skip} to 0 by construction, but because it cannot introduce a missing category its coverage is identical to JointGen's—showing that skip=0 must not be read as constraint satisfaction. EnergyGuide, a single gradient correction without dual updates, barely moves any constraint metric.

Fidelity. The projection stage alone reproduces the trade-off reported for constrained sampling: PCDG-P improves *CategoryTransition*, *DailyLoc* and *Category* on Istanbul but degrades *Distance* (0.1128 → 0.1314) and *Radius* (0.0226 → 0.0516). The full stack removes this cost. On Istanbul *all six* divergences fall below the unconstrained backbone (e.g. Distance −65.0%, Radius −26.5%, DailyLoc −39.9%), and five of six fall below PO-CFG, which was the best raw-fidelity method in the projection-only comparison (G-RANK is effectively tied, 0.1798 vs. 0.1794). On New York four of six improve over the backbone (Distance −76.0%, Radius −66.2%, DailyLoc −26.4%, Category −4.9%).

Residuals we do not hide. On New York, *CategoryTransition* (0.0070 → 0.0183) and *G-RANK* (0.1612 → 0.1914) remain above the unconstrained backbone for every constrained variant of ours, and PO-CFG remains the best method on exactly those two plus *Category*. Enforcing an ordering the data does not exhibit necessarily rewrites adjacent category pairs, so some *CategoryTransition* loss is intrinsic to the task rather than to our mechanism; the *G-RANK* gap is a POI-popularity effect that the distribution correction reduces but does not close on this city (§5.4). All comparisons are between three-seed means; we run no significance tests.

5.3 Ablation of the projection stage

Tables 2 and 3 (in Appendix C, for space) isolate each component by disabling it from the projection stage: No Projection (the backbone), No Existence, No Order, No KL (both KL terms off), No Distance KL (only the distance term off), Fixed Multipliers (no dual update, Eq. (7)), No Gumbel (deterministic relaxation), and Full (= PCDG-P).

Findings. The two energies have cleanly separated roles. Removing the *existence* energy collapses pair coverage to 7.34% (Istanbul) and 10.95% (New York)—*below* the unconstrained backbone's 14.63% and 21.13%—while OVR_{skip} drops to 7.67% and 18.04%: without it, the model satisfies ordering by *omitting* the constrained categories, which is exactly the degeneracy Eq. (4) is designed to block. Removing the *order* energy keeps coverage high (72.53% / 69.92%) but inflates OVR_{strict} from 32.80% / 38.51% to 63.50% / 65.43%. Freezing the multipliers costs 9.90 and 19.94 percentage points of OVR_{strict}, confirming that evaluating the dual update on *hard* violations matters more than the penalty magnitude alone. The *distance-KL* term improves both spatial metrics in both cities (Istanbul Distance 0.1441 → 0.1314, Radius 0.0582 → 0.0516; New York 0.2484 → 0.2210 and 0.1430 → 0.1265) at no measurable constraint cost. The *token-KL* and *Gumbel* terms have small and not uniformly positive marginal effects at this projection budget—removing the Gumbel noise even lowers New York *Distance* to 0.2055—and we report this rather than claiming a benefit for every component.

5.4 Decomposing the fidelity–constraint trade-off

Figure 2 compares the stages in the Unsat–Distance plane; Table 4 in Appendix C gives all stage-wise means.

Where fidelity is lost. Two stages buy constraint satisfaction and both charge for it on the *spatial* axis. The projection (b) raises Istanbul *Radius* from 0.0226 to 0.0516, and category-consistent decoding (c)—which delivers the largest single constraint jump, 32.80% → 3.72% and 38.51% → 9.52% strict violations—raises it further to 0.0892 (0.1265 to 0.1681 on New York). Neither stage degrades the temporal or category-usage marginals; the damage is concentrated in *Distance* and *Radius*.

Where it is recovered. The same-category geometry refinement (d) is constraint-neutral by construction—it swaps only POIs of the same *realized* category together with their catalog coordinates, so every constraint metric is bit-identical to (c) to within 10^{-12}—and it repays the entire spatial cost and more: Istanbul *Distance* 0.1539 → 0.0372 and *Radius* 0.0892 → 0.0159, New York 0.1989 → 0.0610 and 0.1681 → 0.0382. This is the central empirical point: the trade-off is not intrinsic to constrained sampling, it is an artifact of selecting POIs and coordinates without reference to the population spatial distribution. The refinement does, however, concentrate mass on its candidate pools and so degrades the POI-popularity statistic on New York (*G-RANK* 0.1914 → 0.2069). Stage (e) adds the distribution-correction terms and a halved constraint weight, which repairs exactly that statistic (0.2069 → 0.1914; Istanbul 0.1835 → 0.1798) and improves *DailyLoc* in both cities, at the price of a small constraint regression (3.72 → 4.04 and 9.52 → 11.28) and a slightly higher *Distance*.

The loss is spatial, not categorical. The configuration selected from the $\rho \times \lambda \times \beta$ grid (Appendix B) is ρ=0.5, λ=0, β=1: it halves the constraint penalty, applies the POI-distribution term at full weight, and switches the category-side distribution term *off* entirely. Combined with the stage-wise pattern above, this says the residual

Table 1: Method comparison (evaluation v2): fidelity JSDs (upper panel) and constraint metrics in percent (lower panel). Cells are mean ± sample SD over three sampling seeds at fixed checkpoints, not confidence intervals. Bold marks the best unrounded city-wise mean, not significance. PCDG-P: projection; PCDG-PD: adds decoding; PCDG: full stack. New York CategoryTransition and G-RANK remain above JointGen.

Istanbul						
Method	Distance ↓	Radius ↓	CategoryTransition ↓	DailyLoc ↓	Category ↓	G-RANK ↓
JointGen	0.1128 ± 0.0087	0.0226 ± 0.0017	0.0403 ± 0.0001	0.0414 ± 0.0009	0.0649 ± 0.0003	0.1922 ± 0.0022
PostSwap	0.0949 ± 0.0127	0.0226 ± 0.0017	0.0361 ± 0.0005	0.0414 ± 0.0009	0.0647 ± 0.0018	0.1922 ± 0.0022
EnergyGuide	0.1156 ± 0.0052	0.0222 ± 0.0025	0.0424 ± 0.0009	0.0392 ± 0.0020	0.0643 ± 0.0016	0.1921 ± 0.0004
PO-CFG (s=1)	0.0890 ± 0.0050	0.0263 ± 0.0020	0.0365 ± 0.0008	0.0444 ± 0.0022	0.0630 ± 0.0007	**0.1794 ± 0.0026**
PCDG-P (projection)	0.1314 ± 0.0035	0.0516 ± 0.0022	**0.0162 ± 0.0006**	0.0368 ± 0.0014	0.0446 ± 0.0020	0.1820 ± 0.0016
PCDG-PD (+decoding)	0.1539 ± 0.0028	0.0892 ± 0.0024	0.0238 ± 0.0011	0.0297 ± 0.0012	0.0423 ± 0.0009	0.1813 ± 0.0018
PCDG (full)	**0.0395 ± 0.0036**	**0.0166 ± 0.0007**	0.0235 ± 0.0011	**0.0249 ± 0.0009**	**0.0423 ± 0.0018**	0.1798 ± 0.0016
New York						
Method	Distance ↓	Radius ↓	CategoryTransition ↓	DailyLoc ↓	Category ↓	G-RANK ↓
JointGen	0.2904 ± 0.0156	0.1104 ± 0.0071	0.0070 ± 0.0008	0.0409 ± 0.0035	0.0184 ± 0.0002	0.1612 ± 0.0013
PostSwap	0.2276 ± 0.0448	0.1104 ± 0.0071	0.0070 ± 0.0007	0.0409 ± 0.0035	0.0168 ± 0.0006	0.1612 ± 0.0013
EnergyGuide	0.2943 ± 0.0060	0.1112 ± 0.0061	0.0071 ± 0.0011	0.0411 ± 0.0029	0.0183 ± 0.0003	0.1622 ± 0.0009
PO-CFG (s=1)	0.1228 ± 0.0160	0.0823 ± 0.0029	**0.0053 ± 0.0018**	0.0499 ± 0.0008	**0.0121 ± 0.0007**	**0.1525 ± 0.0025**
PCDG-P (projection)	0.2210 ± 0.0224	0.1265 ± 0.0051	0.0132 ± 0.0004	0.0351 ± 0.0035	0.0173 ± 0.0013	0.1942 ± 0.0041
PCDG-PD (+decoding)	0.1989 ± 0.0198	0.1681 ± 0.0042	0.0182 ± 0.0012	0.0335 ± 0.0022	0.0178 ± 0.0012	0.1914 ± 0.0059
PCDG (full)	**0.0698 ± 0.0107**	**0.0373 ± 0.0042**	0.0183 ± 0.0005	**0.0301 ± 0.0010**	0.0175 ± 0.0010	0.1914 ± 0.0060

Istanbul					
Method	OVR_strict ↓	OVR_skip ↓	Pair cov. ↑	Category cov. ↑	Unsat ↓
JointGen	92.89 ± 0.08	48.05 ± 0.74	14.63 ± 0.19	43.19 ± 0.20	95.57 ± 0.15
PostSwap	86.10 ± 0.15	**0.00 ± 0.00**	14.63 ± 0.19	43.19 ± 0.20	85.37 ± 0.19
EnergyGuide	90.66 ± 0.06	44.76 ± 0.58	19.17 ± 0.43	47.06 ± 0.25	92.89 ± 0.16
PO-CFG (s=1)	63.45 ± 0.23	19.09 ± 0.54	43.72 ± 0.35	66.00 ± 0.23	69.07 ± 0.45
PCDG-P (projection)	32.80 ± 0.50	9.12 ± 0.10	70.99 ± 0.08	85.36 ± 0.15	36.86 ± 0.37
PCDG-PD (+decoding)	**3.72 ± 0.13**	1.76 ± 0.10	**94.74 ± 0.16**	**97.81 ± 0.07**	**6.90 ± 0.18**
PCDG (full)	4.04 ± 0.20	2.02 ± 0.15	94.68 ± 0.15	97.75 ± 0.10	7.29 ± 0.20
New York					
Method	OVR_strict ↓	OVR_skip ↓	Pair cov. ↑	Category cov. ↑	Unsat ↓
JointGen	89.45 ± 0.20	52.92 ± 0.86	21.13 ± 0.15	46.75 ± 0.12	93.64 ± 0.26
PostSwap	78.22 ± 0.48	**0.00 ± 0.00**	21.13 ± 0.15	46.75 ± 0.12	78.87 ± 0.15
EnergyGuide	89.17 ± 0.14	51.59 ± 0.88	21.17 ± 0.19	47.14 ± 0.14	93.20 ± 0.39
PO-CFG (s=1)	41.91 ± 1.37	8.97 ± 0.88	59.92 ± 1.58	74.99 ± 1.08	49.48 ± 1.20
PCDG-P (projection)	38.51 ± 0.95	14.11 ± 1.22	67.60 ± 1.86	81.54 ± 0.70	42.42 ± 1.84
PCDG-PD (+decoding)	**9.52 ± 1.04**	3.47 ± 0.64	**88.67 ± 1.74**	**93.92 ± 0.68**	**14.09 ± 2.11**
PCDG (full)	11.28 ± 1.05	4.50 ± 0.70	87.87 ± 1.59	93.46 ± 0.63	15.86 ± 1.90

fidelity cost of partial-order control in this generator lives almost entirely on the spatial/POI side, and that a category-side distribution regularizer is not needed to correct it.

Selection protocol. The 18 configurations were ranked by a fixed rule—maximum, then mean, of the twelve per-city JSD/JointGen ratios—computed on a *training-side* calibration split, and the top-ranked configuration was then evaluated once on the held-out OOD test set. No test metric entered candidate selection, hyper-parameter choice, candidate construction, or the refinement objective. We note explicitly that this calibration split is an engineering calibration rather than an independent validation set (the frozen backbone was trained on those records), and that of the two configurations which reached three-seed confirmation neither cleared all pre-specified calibration thresholds; we report the resulting held-out test numbers as-is rather than re-opening the search.

5.5 Cost and reproducibility

PCDG adds only sampling-time projection iterations to a frozen generator and trains nothing, whereas PO-CFG requires a full retraining per city. We report wall-clock sampling overhead and the number of projection updates in Appendix E, and we release code, constraint matrices, and evaluation scripts to reproduce every table. [TODO: insert sampling-time overhead numbers from distance-v2 diagnostics]

6 Limitations

Several limitations frame our claims honestly. (i) We evaluate two cities; the Istanbul base model was trained historically with a different batch size than New York and its full data provenance is not retroactively verifiable, so we do not claim identical cross-city training conditions. (ii) The reported error bars are sample standard

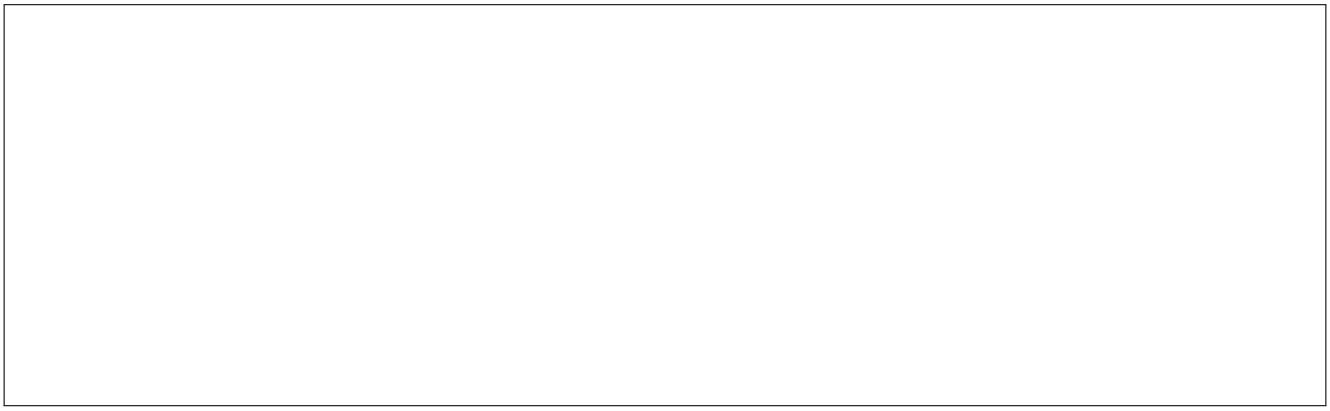


Figure 2: Unsat–Distance trade-off (reserved layout for T2).

deviations over three sampling seeds at a *fixed* checkpoint; they reflect sampling stochasticity, not training variability, and we run no significance tests. (iii) PO-CFG is evaluated at guidance scale 1; we do not claim an extrapolation-tuned optimum. (iv) The distance-KL weight is fixed, not tuned per city, and is a batch-level distribution-matching regularizer rather than a per-sample spatial guarantee. (v) PCDG reduces—but does not certify zero—constraint violations; it reweights the sampling distribution and performs no hard candidate filtering. (vi) **Two New York statistics remain above the unconstrained backbone** (*CategoryTransition* 0.0183 vs. 0.0070; *G-RANK* 0.1914 vs. 0.1612), so our claim is that the trade-off is largely, not entirely, removable; on Istanbul all six divergences improve. (vii) The $\rho/\lambda/\beta$ configuration was selected on a training-side calibration split that the frozen backbone had seen, which is an engineering calibration rather than an independent validation set; no test metric entered selection, but neither of the two three-seed-confirmed configurations cleared all pre-specified calibration thresholds, and we report the held-out test result of the top-ranked configuration as-is. (viii) The transition from the geometry refinement to the full stack changes three things at once—ρ, β, and the composition of the same-category candidate pool—so we do not attribute that difference to any single factor. (ix) The geometry refinement is a migration of an already-frozen configuration; its candidate pools are built from the training reference and the backbone's own scores, but it is not a from-scratch generalization study. (x) We do not yet report a downstream-utility study (e.g., training a next-POI model on the synthetic data). Addressing complex partial-order DAGs (transitivity, conflict detection), adaptive constraint scheduling, a training-time partial-order objective, and downstream utility are natural directions for future work.

7 Conclusion

We studied controllable synthetic generation of POI check-in sequences under sample-level activity-category precedence constraints—a web-native, privacy-relevant problem that single-step conditioning cannot address. We introduced PCDG, a training-free, three-stage sampling-time control stack that compiles a partial-order matrix into differentiable order and existence energies, projects the category-position logits of a frozen joint temporal + category + POI discrete-diffusion generator via a KL-preserving augmented Lagrangian, removes residual category–POI inconsistency by category-consistent decoding, and restores spatial realism through a constraint-neutral, distribution-aware same-category geometry refinement. Across two cities PCDG delivers the strongest constraint satisfaction among post-hoc, guidance, and even fully retrained conditional baselines while *also* improving fidelity over the unconstrained generator on all six divergences on one city and four of six on the other—without retraining anything. Our stage-wise analysis locates the fidelity cost of constrained sampling on the spatial/POI side and shows it is largely an artifact of distribution-blind POI selection rather than a necessary price of control. We hope the training-free, sampling-time recipe proves useful for controllable generation of other structured web content.

A Algorithm and Default Hyper-parameters

Algorithm 1 summarizes one projected reverse step. Defaults: w_{ord}=1, w_{ex}=5, w_{kl}=1, w_{dist}=1; Gumbel temperature T=3; outer iterations 10, inner iterations 50; SGD learning rate η, gradient clip 10; multiplier growth α=2, $\mu_{\max}$=10^3, tolerance τ_{tol}=0.25; distance routes P=8, candidates κ=32, bins 32; applied over the last 40 reverse steps every 4 steps. Geometry refinement (§4.7): w_{d}=1, w_{r}=2, w_{fid}=0.1, P_g=8 paths, κ_g=32 candidates, temperature 1, Adam learning rate 0.05, 200 steps, geometry noise refreshed every 20 steps. Distribution correction (§4.8): ρ=0.5, λ=0, β=1. The geometry step budget was fixed by a speed qualification (median total spatial-sampling time within 1.10× of the unrefined backbone) before any quality comparison, and the weights by the grid of Appendix B. Algorithm 1 covers one projected reverse step only; the decoder of §4.6 acts inside its final line, and the refinement of §4.7 acts once after the reverse process terminates. [TODO: confirm η and step schedule against the distance-v2 manifest before camera-ready.]

B Distribution-Correction Grid and Selection

The distribution-correction stage has three weights. ρ scales the existing constraint loss inside the augmented-Lagrangian objective (Eq. (6)); λ weights a category-side distribution term—the mean of an hour–category JSD and a conditional category-transition JSD against the training reference pool; and β weights a POI-side

Algorithm 1 PCDG projected reverse step (one controlled step s)

Require: backbone logits $\mathbf{z}$; constraints $\{(a, b)\}$ with $\mathbf{W}_A, \mathbf{W}_B$; masks; distance reference H_{train}

$\mathbf{y} \leftarrow \mathbf{z}$ (detached); initialize λ, μ for active rows
for outer = 1 to T_{out} **do**
  for inner = 1 to T_{in} **do**
    relax $\mathbf{y}$ to $\boldsymbol{\pi}$; compute $g^{\text{ord}}, g^{\text{ex}}$ on category positions ▷ Eq. (3)–(4)
    $\mathcal{L} \leftarrow w_{\text{kl}} D_{\text{KL}} + \text{AL}(g^{\text{ord}}, g^{\text{ex}}; \lambda, \mu) + w_{\text{dist}} D_{\text{KL}}(H_{\text{train}} \| H_{\text{proj}})$
    SGD step on $\mathbf{y}$ (clip; restrict to active rows)
  end for
  evaluate hard violations $\hat{\delta}$; update λ, μ ▷ Eq. (7); early stop if satisfied
end for
return joint categorical sample from projected $\mathbf{y}$, decoded category-first ▷ Eq. (9)

distribution term—the mean of an identity-aligned top-1000 visitation JSD and a soft expected-*DailyLoc* JSD—inside the same-category geometry refinement objective. We screened the full grid $\rho \in \{0.25, 0.5, 1\} \times \lambda \in \{0, 1\} \times \beta \in \{0, 0.1, 1\}$ (18 configurations) on a single seed, retained at most two jointly feasible configurations per city, and confirmed those over three seeds, all on a training-side calibration split with 512 conditions per city and seed. Configurations were ranked by the maximum, then the mean, of the twelve per-city JSD/JointGen ratios, with the parameter tuple as the final tie-break. The two configurations that reached confirmation were $(\rho, \lambda, \beta) = (0.5, 0, 1)$ and $(0.5, 0, 0.1)$; the first was selected and is the configuration reported throughout. Spatial results are cached by $(city, pool, seed, \rho, \lambda)$ and reused across the three β refinements of each batch, so β varies at fixed upstream randomness. No test-set metric participated in screening, confirmation, ranking, candidate construction, or the refinement objective. As stated in §6, neither confirmed configuration cleared all pre-specified calibration thresholds: both met the *Distance*, *Radius* and *DailyLoc* ceilings in both cities but not the *CategoryTransition*, *Category* and *G-RANK* ceilings on that calibration split. On the held-out test set the selected configuration then met all eleven gates on Istanbul and nine of eleven on New York. [TODO: Appendix table: all 18 screened configurations with their twelve calibration ratios, from screen-selection.json.]

C Projection and Stage-wise Tables

Tables 2 and 3 are referenced from §5.3 and placed here for space.

D Metric Definitions

[TODO: one-line formal definitions of OVR_strict, OVR_skip, Pair/Category coverage, Unsat, and the six eval-v2 JSD marginals; copy from the unified evaluation module.]

Two definitions that differ from common usage. Radius is computed as $\sqrt{\text{mean}(d(\cdot, \bar{c}))}$, the square root of the mean distance to the arithmetic centroid, *not* the standard root-mean-square radius of gyration; it is therefore comparable across the rows of our tables but not directly against differently defined radii in prior work. *G-RANK* compares the visitation distributions of the top-1000 most frequent POIs; because many POIs tie at the truncation frequency, the membership of that top-1000 set is implementation-dependent under an unstable sort. All tables in this paper use the archived evaluation protocol (v2). We additionally recomputed a deterministic variant that sorts by count descending then POI id ascending and aligns the generated and reference sets by identity before normalization. The effect on *G-RANK* is at most 0.0032 in absolute value and changes no ordering or conclusion: on Istanbul the full stack remains below the backbone (0.1766 vs. 0.1892) and on New York it remains above it (0.1912 vs. 0.1628). All other statistics agree with the archived values to within 10^{-12}.

E Cost

[TODO: sampling wall-clock overhead vs. JointGen, and PO-CFG retraining cost (per-city GPU-hours), from distance-v2 diagnostics.]

References

[1] Jacob Austin, Daniel Johnson, Jonathan Ho, Daniel Tarlow, and Rianne van den Berg. 2021. Structured Denoising Diffusion Models in Discrete State-Spaces. In *Advances in Neural Information Processing Systems*. arXiv preprint arXiv:2107.03006; NeurIPS 2021.

[2] Chu Cao and Mo Li. 2021. Generating Mobility Trajectories with Retained Data Utility. In *Proceedings of the 27th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*. doi:10.1145/3447548.3467158

[3] Michael Cardei, Jacob K. Christopher, Thomas Hartvigsen, Bhavya Kailkhura, and Ferdinando Fioretto. 2025. Constrained Discrete Diffusion. In *Advances in Neural Information Processing Systems 38 (NeurIPS 2025)*. doi:10.52202/085713-0415 39th Conference on Neural Information Processing Systems (NeurIPS 2025).

[4] C. Chen, H. Zhang, P. Wang, and F. Lu. 2024. Simulating Human Mobility with a Trajectory Generation Framework Based on Diffusion Model. *International Journal of Geographical Information Science* (2024). doi:10.1080/13658816.2024.2312199

[5] Jinghong Chen, Weizhe Lin, Jingbiao Mei, and Bill Byrne. 2024. Control-DAG: Constrained Decoding for Non-Autoregressive Directed Acyclic T5 using Weighted Finite State Automata. In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*. doi:10.18653/v1/2024.naacl-short.42

[6] Jialei Chen, Yuanbo Xu, and Yiheng Jiang. 2025. Unlocking the Power of Diffusion Models in Sequential Recommendation: A Simple and Effective Approach. In *Proceedings of the 31st ACM SIGKDD Conference on Knowledge Discovery and Data Mining*. doi:10.1145/3711896.3737172

[7] Bangchao Deng, Ling Ding, Lianhua Ji, Chunhua Chen, Xin Jing, Bingqing Qu, and Dingqi Yang. 2025. Marionette: Fine-Grained Conditional Generative Modeling of Spatiotemporal Human Trajectory Data Beyond Imitation. In *Proceedings of the 31st ACM SIGKDD Conference on Knowledge Discovery and Data Mining*. doi:10.1145/3711896.3737039

[8] Bangchao Deng, Xin Jing, Tianyue Yang, Bingqing Qu, Dingqi Yang, and Philippe Cudre-Mauroux. 2025. Revisiting Synthetic Human Trajectories: Imitative Generation and Benchmarks Beyond Datasaurus. In *Proceedings of the 31st ACM SIGKDD Conference on Knowledge Discovery and Data Mining*. doi:10.1145/3690624.3709180

[9] Jie Feng, Yong Li, Chao Zhang, Funing Sun, Fanchao Meng, Ang Guo, and Depeng Jin. 2018. DeepMove: Predicting Human Mobility with Attentional Recurrent Networks. In *Proceedings of the 2018 World Wide Web Conference*. doi:10.1145/3178876.3186058

[10] Jie Feng, Zeyu Yang, Fengli Xu, Haisu Yu, Mudan Wang, and Yong Li. 2020. Learning to Simulate Human Mobility. In *Proceedings of the 26th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*. doi:10.1145/3394486.3412862

[11] Saibo Geng, Martin Josifoski, Maxime Peyrard, and Robert West. 2023. Grammar-Constrained Decoding for Structured NLP Tasks without Finetuning. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*. doi:10.18653/v1/2023.emnlp-main.674

[12] Letian Gong, Shengnan Guo, Yan Lin, Yichen Liu, Erwen Zheng, Yiwei Shuang, Youfang Lin, Jilin Hu, and Huaiyu Wan. 2025. STCDM: Spatio-Temporal Contrastive Diffusion Model for Check-In Sequence Generation. *IEEE Transactions on Knowledge and Data Engineering* 37, 4 (2025). doi:10.1109/TKDE.2025.3525718

[13] Marta C. González, César A. Hidalgo, and Albert-László Barabási. 2008. Understanding Individual Human Mobility Patterns. *Nature* (2008). doi:10.1038/nature07850

Table 2: Ablation of the projection stage: distributional similarity (JSD ↓, evaluation v2). Full here is PCDG-P, i.e. the projection stage alone.

Istanbul						
Variant	Distance ↓	Radius ↓	CategoryTransition ↓	DailyLoc ↓	Category ↓	G-RANK ↓
No Projection	0.1128 ± 0.0087	0.0226 ± 0.0017	0.0403 ± 0.0001	0.0414 ± 0.0009	0.0649 ± 0.0003	0.1922 ± 0.0022
No Existence	**0.1084 ± 0.0114**	**0.0215 ± 0.0027**	0.0389 ± 0.0005	0.0412 ± 0.0021	0.0661 ± 0.0009	0.1924 ± 0.0028
No Order	0.1176 ± 0.0023	0.0486 ± 0.0035	0.0227 ± 0.0008	**0.0358 ± 0.0006**	0.0439 ± 0.0018	**0.1770 ± 0.0040**
No KL	0.1406 ± 0.0117	0.0574 ± 0.0028	0.0164 ± 0.0006	0.0361 ± 0.0012	**0.0429 ± 0.0012**	0.1817 ± 0.0024
Fixed Multipliers	0.1296 ± 0.0093	0.0526 ± 0.0045	0.0185 ± 0.0005	0.0361 ± 0.0009	0.0457 ± 0.0025	0.1841 ± 0.0020
No Gumbel Noise	0.1315 ± 0.0091	0.0524 ± 0.0023	**0.0160 ± 0.0010**	0.0370 ± 0.0005	0.0439 ± 0.0023	0.1831 ± 0.0027
No Distance KL	0.1441 ± 0.0073	0.0582 ± 0.0025	0.0162 ± 0.0010	0.0366 ± 0.0008	0.0437 ± 0.0023	0.1828 ± 0.0026
Full (=PCDG-P)	0.1314 ± 0.0035	0.0516 ± 0.0022	0.0162 ± 0.0006	0.0368 ± 0.0014	0.0446 ± 0.0020	0.1820 ± 0.0016
New York						
Variant	Distance ↓	Radius ↓	CategoryTransition ↓	DailyLoc ↓	Category ↓	G-RANK ↓
No Projection	0.2904 ± 0.0156	0.1104 ± 0.0071	**0.0070 ± 0.0008**	0.0409 ± 0.0035	0.0184 ± 0.0002	0.1612 ± 0.0013
No Existence	0.2439 ± 0.0277	**0.1004 ± 0.0053**	0.0075 ± 0.0008	0.0393 ± 0.0044	0.0189 ± 0.0004	**0.1587 ± 0.0006**
No Order	0.2175 ± 0.0284	0.1196 ± 0.0108	0.0138 ± 0.0003	0.0352 ± 0.0029	0.0177 ± 0.0003	0.1860 ± 0.0048
No KL	0.2415 ± 0.0239	0.1421 ± 0.0046	0.0125 ± 0.0006	0.0352 ± 0.0041	0.0173 ± 0.0013	0.1939 ± 0.0064
Fixed Multipliers	0.2402 ± 0.0487	0.1218 ± 0.0018	0.0127 ± 0.0005	0.0353 ± 0.0041	0.0175 ± 0.0006	0.1891 ± 0.0023
No Gumbel Noise	**0.2055 ± 0.0205**	0.1248 ± 0.0056	0.0130 ± 0.0010	0.0360 ± 0.0036	0.0173 ± 0.0003	0.1918 ± 0.0041
No Distance KL	0.2484 ± 0.0267	0.1430 ± 0.0055	0.0134 ± 0.0011	0.0355 ± 0.0031	**0.0172 ± 0.0013**	0.1937 ± 0.0082
Full (=PCDG-P)	0.2210 ± 0.0224	0.1265 ± 0.0051	0.0132 ± 0.0004	**0.0351 ± 0.0035**	0.0173 ± 0.0013	0.1942 ± 0.0041

Mean ± sample standard deviation over three sampling seeds. Bold denotes the best unrounded mean within each city, not statistical significance. No KL disables both the token-KL anchor and the distance-KL term; No Distance KL disables only the latter. Removing the KL or Gumbel terms does not uniformly degrade fidelity, which we report rather than claiming a benefit for every component.

Table 3: Ablation of the projection stage: constraint satisfaction (%, evaluation v2). The existence energy drives coverage, the order energy drives strict ordering, and the adaptive multipliers beat a fixed penalty.

Istanbul					
Variant	OVR_strict ↓	OVR_skip ↓	Pair cov. ↑	Category cov. ↑	Unsat ↓
No Projection	92.89 ± 0.08	48.05 ± 0.74	14.63 ± 0.19	43.19 ± 0.20	95.57 ± 0.15
No Existence	93.67 ± 0.15	**7.67 ± 0.56**	7.34 ± 0.12	36.92 ± 0.19	93.58 ± 0.17
No Order	63.50 ± 0.76	50.34 ± 0.73	**72.53 ± 0.70**	**86.08 ± 0.17**	70.70 ± 0.73
No KL	32.85 ± 0.52	9.20 ± 0.29	71.11 ± 0.08	85.40 ± 0.16	36.75 ± 0.57
Fixed Multipliers	42.70 ± 0.51	20.97 ± 0.90	71.25 ± 0.89	85.00 ± 0.42	45.64 ± 0.58
No Gumbel Noise	**32.75 ± 0.39**	9.13 ± 0.22	70.97 ± 0.34	85.38 ± 0.19	**36.75 ± 0.25**
No Distance KL	32.96 ± 0.46	9.10 ± 0.16	70.92 ± 0.23	85.29 ± 0.14	36.83 ± 0.24
Full (=PCDG-P)	32.80 ± 0.50	9.12 ± 0.10	70.99 ± 0.08	85.36 ± 0.15	36.86 ± 0.37
New York					
Variant	OVR_strict ↓	OVR_skip ↓	Pair cov. ↑	Category cov. ↑	Unsat ↓
No Projection	89.45 ± 0.20	52.92 ± 0.86	21.13 ± 0.15	46.75 ± 0.12	93.64 ± 0.26
No Existence	90.38 ± 0.47	18.04 ± 1.26	10.95 ± 0.45	39.47 ± 0.12	91.80 ± 0.40
No Order	65.43 ± 0.12	51.24 ± 0.30	**69.92 ± 0.74**	**82.84 ± 0.40**	72.86 ± 0.94
No KL	**38.24 ± 0.96**	13.98 ± 0.82	67.67 ± 1.56	81.57 ± 0.72	**42.40 ± 1.48**
Fixed Multipliers	58.45 ± 0.66	35.14 ± 0.60	63.79 ± 0.76	78.37 ± 0.35	60.67 ± 0.49
No Gumbel Noise	39.11 ± 1.53	13.93 ± 1.34	66.41 ± 1.32	80.95 ± 0.65	43.91 ± 1.59
No Distance KL	38.48 ± 1.04	**13.76 ± 0.72**	67.33 ± 1.86	81.38 ± 0.81	42.77 ± 1.79
Full (=PCDG-P)	38.51 ± 0.95	14.11 ± 1.22	67.60 ± 1.86	81.54 ± 0.70	42.42 ± 1.84

Mean ± sample standard deviation over three sampling seeds. Bold denotes the best unrounded mean within each city. Removing the existence energy collapses coverage *below* the unconstrained backbone, confirming that ordering can otherwise be satisfied by omitting the constrained categories.

[14] A. Haydari, D. Chen, Z. Lai, et al. 2026. MobilityGPT: Enhanced Human Mobility Modeling With a GPT Model. *IEEE Transactions on Intelligent Transportation Systems* (2026). doi:10.1109/TITS.2025.3626357

[15] Jonathan Ho, Ajay Jain, and Pieter Abbeel. 2020. Denoising Diffusion Probabilistic Models. In *Advances in Neural Information Processing Systems*. arXiv preprint arXiv:2006.11239; NeurIPS 2020.

[16] S. L. Hsu, E. Tung, J. Krumm, et al. 2024. TrajGPT: Controlled Synthetic Trajectory Generation Using a Multitask Transformer-Based Spatiotemporal Model. In *Proceedings of the 32nd ACM International Conference on Advances in Geographic Information Systems*. doi:10.1145/3678717.3691303

[17] Y. Hu, Y. Du, Z. Zhang, et al. 2024. Real-Time Trajectory Synthesis with Local Differential Privacy. In *Proceedings of the IEEE International Conference on Data Engineering*. doi:10.1109/ICDE60146.2024.00137

[18] Yanmei Hu, Minghao Yin, and Dunbo Cai. 2011. On the Discovery and Utility of Precedence Constraints in Temporal Planning. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 25. doi:10.1609/aaai.v25i1.8052

[19] W. Jiang, W. X. Zhao, J. Wang, and J. Jiang. 2023. Continuous Trajectory Generation Based on Two-Stage GAN. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 37. doi:10.1609/aaai.v37i4.25557

[20] S. Li, T. Tran, H. Lin, et al. 2025. Geo-Llama: Leveraging LLMs for Human Mobility Trajectory Generation with Constraints. In *Proceedings of the 2025 IEEE International Conference on Mobile Data Management*. doi:10.1109/MDM65600.2025.00023

[21] H. Lin, S. Shaham, Y. Y. Chiang, and C. Shahabi. 2023. Generating Realistic and Representative Trajectories with Mobility Behavior Clustering. In *Proceedings of the 31st ACM International Conference on Advances in Geographic Information*

Table 4: Stage-wise comparison, evaluation-v2 three-seed means. SDs for (a), (b), (c), and (e) are in Table 1. Geometry (d) preserves (c)'s constraints within 10^{-12} and uses a previously fixed configuration. Transition (d)→(e) jointly changes ρ $(1 \to 0.5)$, β $(0 \to 1)$, and the same-category candidate pool; it does not isolate a single factor. The calibration grid (Appendix B) holds the pool construction fixed.

	Distributional similarity (JSD ↓)						Constraints (%)		
Stage	Distance	Radius	CatTrans	DailyLoc	Category	G-RANK	OVR_str. ↓	Pair cov. ↑	Unsat ↓
Istanbul									
(a) JointGen (no control)	0.1128	0.0226	0.0403	0.0414	0.0649	0.1922	92.89	14.63	95.57
(b) + order/existence projection	0.1314	0.0516	0.0162	0.0368	0.0446	0.1820	32.80	70.99	36.86
(c) + category-consistent decoding	0.1539	0.0892	0.0238	0.0297	0.0423	0.1813	3.72	94.74	6.90
(d) + geometry refinement	0.0372	0.0159	0.0238	0.0284	0.0423	0.1835	3.72	94.74	6.90
(e) + distribution correction (full)	0.0395	0.0166	0.0235	0.0249	0.0423	0.1798	4.04	94.68	7.29
New York									
(a) JointGen (no control)	0.2904	0.1104	0.0070	0.0409	0.0184	0.1612	89.45	21.13	93.64
(b) + order/existence projection	0.2210	0.1265	0.0132	0.0351	0.0173	0.1942	38.51	67.60	42.42
(c) + category-consistent decoding	0.1989	0.1681	0.0182	0.0335	0.0178	0.1914	9.52	88.67	14.09
(d) + geometry refinement	0.0610	0.0382	0.0182	0.0335	0.0178	0.2069	9.52	88.67	14.09
(e) + distribution correction (full)	0.0698	0.0373	0.0183	0.0301	0.0175	0.1914	11.28	87.87	15.86

Systems. doi:10.1145/3589132.3625657

[22] Jing Long, Guanhua Ye, Tong Chen, Yang Wang, Meng Wang, and Hongzhi Yin. 2024. Diffusion-Based Cloud-Edge-Device Collaborative Learning for Next POI Recommendations. In *Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*. doi:10.1145/3637528.3671743

[23] Q. Long, C. Rong, T. Li, and Y. Li. 2026. Dynamic Population Distribution Aware Human Trajectory Generation with Diffusion Model. *ACM Transactions on Intelligent Systems and Technology* (2026). doi:10.1145/3787459

[24] Qingyue Long, Huandong Wang, Tong Li, Lisi Huang, Kun Wang, Qiong Wu, Guangyu Li, Yanping Liang, Li Yu, and Yong Li. 2023. Practical Synthetic Human Trajectories Generation Based on Variational Point Processes. In *Proceedings of the 29th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*. doi:10.1145/3580305.3599888

[25] Aaron Lou, Chenlin Meng, and Stefano Ermon. 2024. Discrete Diffusion Modeling by Estimating the Ratios of the Data Distribution. In *Proceedings of the 41st International Conference on Machine Learning*. arXiv preprint arXiv:2310.16834; ICML 2024 Oral.

[26] Yingtao Luo, Qiang Liu, and Zhaocheng Liu. 2021. STAN: Spatio-Temporal Attention Network for Next Location Recommendation. In *Proceedings of the Web Conference 2021*. doi:10.1145/3442381.3449998

[27] Chenlin Meng, Kristy Choi, Jiaming Song, and Stefano Ermon. 2022. Concrete Score Matching: Generalized Score Matching for Discrete Data. In *Advances in Neural Information Processing Systems*, Vol. 35. doi:10.52202/068431-2502

[28] F. Netzler and M. Lienkamp. 2024. Privacy Preserving Human Mobility Generation Using Grid-Based Data and Graph Autoencoders. *ISPRS International Journal of Geo-Information* 13, 7 (2024). doi:10.3390/ijgi13070245

[29] Y. Qin, H. Wu, W. Ju, et al. 2023. A Diffusion Model for POI Recommendation. *ACM Transactions on Information Systems* (2023). doi:10.1145/3624475

[30] J. Rao, S. Gao, and S. Zhu. 2023. CATS: Conditional Adversarial Trajectory Synthesis for Privacy-Preserving Trajectory Data Publication Using Deep Learning Approaches. *International Journal of Geographical Information Science* (2023). doi:10.1080/13658816.2023.2262550

[31] X. Rao, S. Shang, R. Jiang, et al. 2025. Seed: Bridging Sequence and Diffusion Models for Road Trajectory Generation. In *Proceedings of the ACM Web Conference 2025*. doi:10.1145/3696410.3714951

[32] Subham Sahoo, Marianne Arriola, Yair Schiff, Aaron Gokaslan, Edgar Marroquin, Justin Chiu, Alexander Rush, and Volodymyr Kuleshov. 2024. Simple and Effective Masked Diffusion Language Models. In *Advances in Neural Information Processing Systems*, Vol. 37. doi:10.52202/079017-4135

[33] Yair Schiff, Subham Sekhar Sahoo, Hao Phung, Guanghan Wang, Sam Boshar, Hugo Dalla-Torre, Bernardo P. de Almeida, Alexander Rush, Thomas Pierrot, and Volodymyr Kuleshov. 2025. Simple Guidance Mechanisms for Discrete Diffusion Models. In *International Conference on Learning Representations*. arXiv preprint arXiv:2412.10193; ICLR 2025.

[34] Chaoming Song, Zehui Qu, Nicholas Blumm, and Albert-László Barabási. 2010. Limits of Predictability in Human Mobility. *Science* (2010). doi:10.1126/science.1177170

[35] X. Sun, Q. Ye, H. Hu, et al. 2023. Synthesizing Realistic Trajectory Data With Differential Privacy. *IEEE Transactions on Intelligent Transportation Systems* (2023). doi:10.1109/TITS.2023.3241290

[36] D. Wang, Y. Huang, S. Gao, et al. 2025. Generative Next POI Recommendation with Semantic ID. In *Proceedings of the 31st ACM SIGKDD Conference on Knowledge Discovery and Data Mining*. doi:10.1145/3711896.3736981

[37] H. Wang, C. Gao, Y. Wu, et al. 2023. PateGail: A Privacy-Preserving Mobility Trajectory Generator with Imitation Learning. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 37. doi:10.1609/aaai.v37i12.26700

[38] Ziwei Wang, Jun Zeng, Lin Zhong, Ling Liu, Min Gao, and Junhao Wen. 2024. DS-DRec: Next POI Recommendation Using Deep Semantic Extraction and Diffusion Model. *Information Sciences* (2024). doi:10.1016/j.ins.2024.121004

[39] T. Wei, Y. Lin, S. Guo, et al. 2024. Diff-RNTraj: A Structure-Aware Diffusion Model for Road Network-Constrained Trajectory Generation. *IEEE Transactions on Knowledge and Data Engineering* (2024). doi:10.1109/TKDE.2024.3460051

[40] G. Xiong, Z. Li, M. Zhao, et al. 2024. TrajSGAN: A Semantic-Guiding Adversarial Network for Urban Trajectory Generation. *IEEE Transactions on Computational Social Systems* (2024). doi:10.1109/TCSS.2023.3235923

[41] Song Yang, Jiamou Liu, and Kaiqi Zhao. 2022. GETNext: Trajectory Flow Map Enhanced Transformer for Next POI Recommendation. In *Proceedings of the 45th International ACM SIGIR Conference on Research and Development in Information Retrieval*. doi:10.1145/3477495.3531983

[42] Yuan Yuan, Jingtao Ding, Chenyang Shao, Depeng Jin, and Yong Li. 2023. Spatio-temporal Diffusion Point Processes. In *Proceedings of the 29th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*. doi:10.1145/3580305.3599511

[43] Yuan Yuan, Jingtao Ding, Huandong Wang, Depeng Jin, and Yong Li. 2022. Activity Trajectory Generation via Modeling Spatiotemporal Dynamics. In *Proceedings of the 28th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*. doi:10.1145/3534678.3542671

[44] K. Zhang, Y. Pang, Y. Zhang, and Y. Sekimoto. 2024. MobGLM: A Large Language Model for Synthetic Human Mobility Generation. In *Proceedings of the 32nd ACM International Conference on Advances in Geographic Information Systems*. doi:10.1145/3678717.3691311

[45] Yuanshao Zhu, Yongchao Ye, Shiyao Zhang, Xiangyu Zhao, and James J. Q. Yu. 2023. DiffTraj: Generating GPS Trajectory with Diffusion Probabilistic Model. In *Advances in Neural Information Processing Systems*, Vol. 36. doi:10.52202/075280-2843

[46] Yuanshao Zhu, James Jianqiao Yu, Xiangyu Zhao, Qidong Liu, Yongchao Ye, Wei Chen, Zijian Zhang, Xuetao Wei, and Yuxuan Liang. 2024. ControlTraj: Controllable Trajectory Generation with Topology-Constrained Diffusion Model. In *Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*. doi:10.1145/3637528.3671866